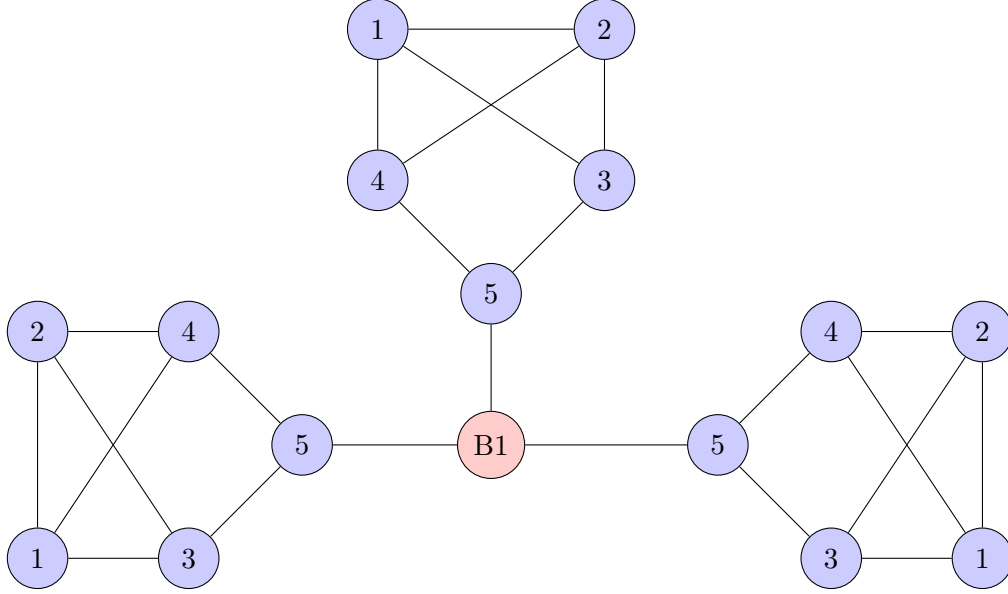


Question 1: Find a cubic graph which does not contain a 1-factor

By Question 2, we need at least 3 bridges. The idea is that our bridges should not be independent, as if they were, our scenario is similar as to if we had 2 bridges.



To prove this does not contain a 1-factor, consider removing vertex $B1$. This results in a disconnected graph with 3 odd components, but we removed 1 vertex. Thus Tutte's Theorem tells us that this graph does not contain a 1-factor.

Question 2: Prove that every cubic graph with at most two bridges contains a 1-factor.

It is sufficient to show that every cubic graph with at most 2 bridges satisfies Tutte's condition. Let $S \subseteq V(G)$ and let C be an odd component of $G - S$. The sum of degree (in G) of vertices in C must be odd, since C is cubic. But by the Handshaking Lemma, only an even part of this sum arises from edges contained in C . Thus there is an odd sum joining C to S and thus there is an odd number of edges. Now this means, for each odd component C , there can be 1 edge joining C to S (corresponding to a bridge) or at least 3 edges. Let r be the number of odd components joining to S with only 1 edge. Then there are at least $3(q(G - S) - r) + r = 3q(G - S) - 2r$ edges from odd components joining S . Clearly there are at most $3|S|$ edges joining odd components from S . So

$$3q(G - S) - 2r \leq 3|S| \implies q(G - S) \leq |S| + \frac{2}{3}r$$

Now, $r \leq 2$ and $q(G - S)$ and $|S|$ are integers, so we have a stricter inequality:

$$q(G - S) \leq |S| + \frac{4}{3} \implies q(G - S) \leq |S| + 1$$

Assume now that $q(G - S) = |S| + 1$. Then from our initial inequality,

$$|S| + 1 \leq |S| + \frac{2}{3}r \iff r \geq \frac{3}{2}$$

Again, r is an integer and so $r = 2$. But then G has 2 bridges and they join different odd components to S . Our lower bound for edges from odd components to S is then $3(|S| + 1) - 4 = 3|S| - 1$. This must then be attained (the other case implies the number of edges is a multiple of 3, but the 2 bridges mean it cannot be), and every other odd component joins S with 3 edges (as if any odd component joined with more edges, they would be at least 5 edges and contradict our upper bound). So there is one edge that does not connect S to an odd vertex, and thus either is internal in S or joins to an even component. In both cases, this edge is a bridge which contradicts that G has at most 2 bridges. Thus $q(G - S) = |S| + 1$ is impossible and $q(G - S) \leq |S|$, so the Marriage Condition is satisfied.

Question 3: Provide a proof of the Marriage Theorem which utilises Tutte's Theorem.

Theorem 0.1 (Hall's Marriage Theorem). Let $G = (X, Y, E)$ be a finite bipartite graph with bipartite sets X and Y . Then G contains a matching that covers every vertex of X if and only if for every subset $W \subseteq X$, we have

$$|N_G(W)| \geq |W|,$$

where $N_G(W)$ denotes the set of all neighbors in Y of vertices in W .

Suppose G contains a matching that covers every vertex of X . Then the subgraph H , induced by X , and the set of vertices in Y that are used, contains a perfect matching. But then by Tutte's Theorem, for every $S \subseteq V(H)$, we have $q(H - S) \leq |S|$. Let $W \subseteq X$. Then $q(H - N_H(W)) \leq |N_H(W)| \leq |N_G(W)|$. In $H - N_H(W)$, every vertex in W is isolated (and thus an odd component). Finally,

$$|W| \leq q(H - N_H(W)) \leq |N_G(W)|$$

The other direction is proved in Theorem 6.2.4. in the notes.

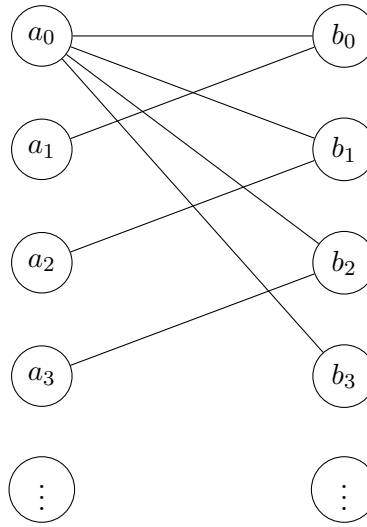
Question 4: Let M be a sub-optimal matching of a bipartite graph G (M contains less edges than some other matching of G). Prove that G contains an augmenting path with respect to M . Does this fact generalise to non-bipartite graphs? (Hint: use symmetric difference).

Let M be a sub-optimal matching, containing less edges than matching N . Then the symmetric difference, $M \triangle N = (M - N) \cup (N - M)$ is the edges in either M or N , but not both. This is necessarily nonempty because $|N| > |M|$. Now consider the connected components of $G[M \triangle N]$. Every vertex has degree at most 2 (if they are matched by both M and N , less if they are only matched by one). Thus the connected components are either even cycles (alternating edges from M and N , equal number from both in each cycle), or paths, possibly with some unmatched endpoints. Since $|N| > |M|$, there must be at least one path component in $M \triangle N$ with more edges from N than from M . Such a path, necessarily of odd length, starts and ends at vertices unmatched by M and consists of alternating edges from N and M and is thus

an augmenting path for M . This fact does generalise to non-bipartite graphs but is complicated (Berge's Theorem)

Question 5: Find an infinite graph which serves as a counter-example to the Marriage Theorem.

Let $A = \{a_0, a_1, \dots\}, B = \{b_0, b_1, \dots\}$ with edges a_0b_i for every $i \in \mathbb{N}_0$, and a_jb_{j-1} for every $j \in \mathbb{N}^+$. We cannot find a matching that covers X , as if it did, every edge a_jb_{j-1} must be included, and any edge incident to a_0 provides a contradiction. But, for $W \subseteq X$, if $a_0 \notin W$, $|W| = |N_G(W)|$. If $a_0 \in W$, then $|N_G(W)| = \aleph_0 \geq |W|$.



Question 6: Show that a graph G is bipartite if and only if $\beta(H) \geq \frac{1}{2}|V(H)|$ for every subgraph H of G

Suppose G is bipartite and H is a subgraph of G . Let $\{A, B\}$ be a bipartition of G . Then $\{A_H, B_H\} := \{A \cap V(H), B \cap V(H)\}$ is a bipartition of H , and each set is independent. At least one of these sets must have a size $\geq \frac{1}{2}|V(H)|$, as $V(H) = A_H \cup B_H$. But then $\beta(H) \geq \max\{|A_H|, |B_H|\} \geq \frac{1}{2}|V(H)|$. Now suppose that G is not bipartite. Then G contains an odd cycle $H := v_1v_2\dots v_nv_1$. Then $V(H) = n$. The maximum number of independent vertices is $\frac{n-1}{2} < \frac{1}{2}n$.

Question 7: Show that every tree has at most one perfect matching

Let T be a tree and M_1 and M_2 be 2 distinct perfect matchings of T . Let $v_1v_2 \in M_1$ with $v_1v_2 \notin M_2$. Then there must exist an edge incident to v_2 in M_2 . Let this be v_2v_3 . We cannot have that $v_3 = v_1$. Continuing this process, we get alternating edges $v_3v_4 \in M_1, v_4v_5 \in M_2, \dots, v_kv_{k+1} \in M_1, v_{k+1}v_{k+2} \in M_2$ and so on. If any $v_i = v_j$, $i \neq j$, this implies the existence of a cycle, which cannot be. Thus the alternating path terminates, which is also a contradiction. Thus every tree has at most one perfect matching.

Question 8: Prove or disprove: A graph G without isolated vertices has a perfect matching if and only if $\alpha'(G) = \beta'(G)$

This is true. First suppose G has a perfect matching M . Then $|M| = n/2$ and so $\beta'(G) = |M| = n/2$ (as a perfect matching is also a maximum matching). But we know that a graph with no isolated vertices satisfies $\alpha'(G) + \beta'(G) = n$. So $\alpha'(G) = n/2 = \beta'(G)$. On the other hand, suppose $\alpha'(G) = \beta'(G)$. Again, this implies $\beta'(G) = n/2$ and we can find a matching M with $n/2$ vertices. This must then be a perfect matching, as each edge is independent and covers 2 vertices.

Question 9: Show that if G is a bipartite graph without isolated vertices, then $\alpha'(G) = \beta(G)$

We know that

$$\begin{aligned}\alpha(G) + \beta(G) &= n \\ \alpha'(G) + \beta'(G) &= n \\ \alpha(G) &= \beta'(G)\end{aligned}$$

(all proven in the notes). This system of inequalities implies $\alpha'(G) = \beta(G)$.